

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (VII)	Branch: Computer Science & Engineering
Subject: Machine Learning	Course Code: D022711(022)
Total / Minimum-Pass Marks (End Semester Exam): 100 / 35	L: 3 T: 1 P: 0 Credits: 4
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Course Objectives

1. To introduce the basic concepts of Machine Learning.
2. To introduce the concepts of Data Analysis in Machine Learning.
3. To introduce the concepts of Machine Learning algorithms.
4. To introduce the concepts related to Machine Learning model diagnosis and tuning.
5. To introduce the concepts related to Artificial Neural Networks.

UNIT- I Introduction: History and Evolution, Machine Learning Categories: Supervised Learning, Unsupervised Learning, Reinforcement Learning. Knowledge Discovery in Databases, SEMMA (Sample, Explore, Modify, Model, Assess).

UNIT-II Machine Learning Perspective of Data: Scales of Measurement, Dealing with Missing Data, Handling Categorical Data, Normalizing Data, Feature Construction or Generation. Correlation and Causation, Polynomial Regression, Logistic Regression, ROC Curve.

UNIT- III Introduction to Machine Learning Algorithms: Decision Trees, Support Vector Machine, k-Nearest Neighbors, Time-Series Forecasting, Clustering, Principal Component Analysis (PCA).

UNIT-IV Model Diagnosis and Tuning: Bias and Variance, K-Fold Cross Validation, Bagging, RandomForest, Gradient Boosting, Stacking.

UNIT-V Artificial Neural Network (ANN): Perceptron—Single Artificial Neuron, Multilayer Perceptrons (Feedforward Neural Network), Restricted Boltzman Machines (RBMs).

Text Books:

1. Mastering Machine Learning with Python in Six Steps– Manohar Swamynathan, APress.
2. Python Machine Learning for Beginners, M. Usman Malik, AI Publishing.

Reference Books:

1. Introduction to Machine Learning with Python, Daniel Nedal& Peters Morgan, AI Sciences
2. Data Structures and Algorithms Using Python, Rance D. Necaise, WILEY

Course Outcomes [After undergoing the course, students will be able to:]

1. Remember the basic concepts of Machine Learning.
2. Able to perform Data Analysis in Machine Learning.
3. Remember the Machine Learning algorithms.
4. Able to perform Machine Learning model diagnosis and tuning.
5. Apply the concept of Artificial Neural Networks.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (VII)	Branch: Computer Science & Engineering
Subject: Data mining and Warehousing	Course Code: D022712(022)
Total / Minimum-Pass Marks (End Semester Exam): 100 / 35	L: 2 T: 1 P: 0 Credits: 3
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Course Objectives

1. To understand the overall architecture of a data warehouse.
2. The different data mining models and techniques will be discussed in this course.
3. Evaluate different models used for OLAP and data pre-processing;
4. Design and implement systems for data mining and evaluate the performance of different data mining algorithms;
5. Propose data mining solutions for different applications.
6. Differentiate Online Transaction Processing and Online Analytical processing

UNIT-I Overview and Concepts: Need for data warehousing, basic elements of data warehousing, Trends in data warehousing. Planning and Requirements: Project planning and management, Collecting the requirements. Architecture And Infrastructure: Architectural components, Infrastructure and metadata.

UNIT-II Data Design and Data Representation: Principles of dimensional modelling, Dimensional modelling advanced topics, data extraction, transformation and loading, data quality

UNIT-III Information Access and Delivery: Matching information to classes of users, OLAP in data warehouse, Data warehousing and the web. Implementation And Maintenance: Physical design process, data warehouse deployment, growth and maintenance.

UNIT-IV Data Mining: Introduction: Basics of data mining, related concepts, Data mining techniques Data Mining Algorithms: Classification, Clustering, Association rules. Knowledge Discovery: KDD Process.

UNIT-V Web Mining: Web Content Mining, Web Structure Mining, Web Usage mining. Advanced Topics: Spatial mining, Temporal mining. Visualization : Data generalization and summarization-based characterization, Analytical characterization: analysis of attribute relevance, Mining class comparisons: Discriminating between different classes, Mining descriptive statistical measures in large databases Data Mining Primitives, Languages, and System Architectures: Data mining Primitives, Query language, Designing GUI based on a data mining query language, Architectures of data mining systems Application and Trends in Data Mining: Applications, Systems products and research prototypes, Additional themes in data mining, Trends in data mining.

Text Books:

1. Data warehousing- concepts, Techniques, Products and Applications by Prabhu, Prentice hall of India
2. Insight into Data Mining: Theory & Practice by Soman K P, Prentice hall of India.
3. Data Mining Introductory and Advanced Topics by M.H. Dunham, Pearson Education.

Reference Books:

1. Data Warehousing Fundamentals by PaulrajPonniah, John Wiley.
2. Introduction to Data mining with Case Studies by Gupta, PHI.
3. The Data Warehouse Lifecycle toolkit by Ralph Kimball, John Wiley.
4. Introduction to Building the Data warehouse, IBM, PHI.

Course Outcome: After successful completion of this course students will be able to

1. Design a data warehouse for an organization
2. Develop skills to write queries using DMQL
3. Extract knowledge using data mining techniques
4. Adapt to new data mining tools.
5. Explore recent trends in data mining such as web mining, spatial-temporal mining.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (VII)	Branch: Computer Science & Engineering
Subject: Internet and Web Technology	Course Code: D022713(022)
Total / Minimum-Pass Marks (End Semester Exam): 100 / 35	L: 2 T: 1 P: 0 Credits: 3
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Course Objectives

1. Describe the important features of the Web and Web browser software
2. Evaluate e-mail software and Web-based e-mail services
3. Use FTP and other services to transfer and store data
4. Demonstrate the use of real-time chat and briefly describe the history of the wireless Internet
5. Create HTML documents and enhance them with browser extensions

UNIT-I INTRODUCTION TO INTERNET: Introduction, Evolution of Internet, Internet Applications, Internet Protocol -TCP/IP, UDP, HTTP, Secure Http(Shttp) Internet Addressing – Addressing Scheme – Ipv4 & IPv6, Network Byte Order, Domain Name Server and IP Addresses, Mapping . Internet Service Providers, Types Of Connectivity Such As Dial-Up Leaded Vsat Etc. Web Technologies: ThreeTier Web Based Architecture; Jsp, Asp, J2ee, .Net Systems

UNIT-II HTML CSS AND SCRIPTING: HTML - Introduction, Sgml, Dtd(Document Type Definition, Basic Html Elements, Tags and usages, HTML Standards , Issues in HTML Dhtml: Introduction Cascading Style Sheets: Syntax ,Class Selector, Id Selector Dom (Document ObjectModel) &Dso (Data Source Object) Approaches To Dynamic Pages: Cgi, Java Applets, Plug Ins, Active X, Java Script –Java Script Object Model, Variables-Constant – Expressions, Conditions-Relational Operators- Data Types – FlowControl – Functions & Objects-events and event handlers – Data type Conversion & Equality – Accessing HTML form elements

UNIT-III XML: What is XML – Basic Standards, Schema Standards, Linking & Presentation Standards, Standards that build on XML, Generating XML data, Writing a simple XML File, Creating a Document type definition, Documents &Data ,DefiningAttributes & Entities in the DTD ,Defining Parameter Entities & conditional Sections, Resolving a naming conflict, UsingNamespaces, Designing an XML data structure, Normalizing Data, Normalizing DTDS

UNIT-IV INTERNET SECURITY & FIREWALLS: Security Threats From Mobile Codes, Types Of Viruses, Client Server Security Threats, Data & Message Security, Various electronic payment systems, Introduction to EDI, Challenges– Response System, Encrypted Documents And Emails,Firewalls: Hardened Firewall Hosts, Ip- Packet Screening, Proxy Application Gateways, Aaa (Authentication, AuthorizationAnd Accounting).

UNIT-V WEBSITE PLANNING & HOSTING: Introduction, Web Page Lay-Outing, Where To Host Site, Maintenance Of Site, Registration Of Site On Search Engines And Indexes, Introduction To File Transfer Protocol, Public Domain Software, Types Of Ftp Servers (Including Anonymous),Ftp Clients Common Command. Telnet Protocol, Server Domain, Telnet Client, Terminal Emulation. Usenet And Internet Relay Chat

Text Books:

1. Internet & Intranet Engineering,- Daniel Minoli, TMH.
2. Alexis Leon and Mathews Leon – Internet for Every One, Tech World.

Reference Books:

1. Eric Ladd, Jim O'Donnel–“Using HTML 4, XML and JAVA”-Prentice Hall of India -1999.
2. “Beginning Java Script “– Paul Wilton – SPD Publications –2001.
3. Frontiers of Electronics of Commerce, Ravi kalakota& Andrew B. Whinston Addison Wesley

Course Outcomes [After undergoing the course, students will be able to:]

1. Understand, analyze and apply the role of languages like HTML, DHTML, CSS, XML, Javascript, and web applications
2. Analyze a web page and identify its elements and attributes.
3. Create XML documents and XML Schema

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (VII)	Branch: Computer Science & Engineering
Subject: Parallel and Distributed Algorithms (Professional Elective III)	Course Code: D022731(022)
Total / Minimum-Pass Marks (End Semester Exam): 100 / 35	L: 1 T: 1 P: 0 Credits: 2
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Parallel Algorithms

Unit-I: Introduction Parallel Programming Models: Shared-memory model (PRAM, MIMD, SIMD), network model (line, ring, mesh, hypercube), performance measurement of parallel algorithms.

Unit-II Algorithm Design Techniques for PRAM Models: Balancing, divide and conquer, parallel prefix computation, pointer jumping, symmetry breaking, pipelining, accelerated cascading. Algorithms for PRAM Models: Parallel Reduction, Prefix Sums, List ranking, sorting and searching, tree algorithms, graph algorithms, string algorithms.

Unit-III Algorithms for Network Models: Matrix algorithms, sorting, graph algorithms, routing, Relationship with PRAM models. Parallel Complexity: Lower bounds for PRAM models, the complexity class NC, P-completeness.

Distributed Algorithms

Unit-IV Basic concepts. Models of computation: shared memory and message passing systems, synchronous and asynchronous systems. Logical time and event ordering. Global state and snapshot algorithms, clock synchronization. Distributed Operating Systems: Mutual exclusion, deadlock detection

Unit-V Classical Algorithms: Leader election, termination detection, distributed graph algorithms. Fault tolerance and recovery: basic concepts, fault models, agreement problems and its applications, commit protocols, voting protocols, check-pointing and recovery, reliable communication. Security and Authentication: basic concepts, Kerberos. Resource sharing and load balancing.

Text Books:

1. Joseph F Jája, An Introduction to Parallel Algorithms, Addison-Wesley, 1992.
2. Joseph Jaja, An Introduction to Parallel Algorithms, Addison Wesley
3. Mukesh Singhal and Niranjan Shivaratri, Advanced Concepts in Operating Systems, McGraw-Hill.

Reference Books:

1. Michael J Quinn, Parallel Computing: Theory and Practice, second edition, McGraw Hill, 1994/2002.
2. Michael J Quinn, Parallel Programming in C with MPI and OpenMP, first edition, McGraw Hill, 2004/2003.
3. AnanthGrama, Anshul Gupta, George Karypis and Vipin Kumar, Introduction to Parallel Computing, second edition, Addison-Wesley/Pearson, 1994/2003.
4. Nancy Lynch, Distributed Algorithms, Morgan Kaufmann.
5. Andrew S. Tanenbaum, Distributed Operating Systems, ACM Press.
6. Jie Wu, Distributed Systems, CRC Press.
7. HagitAttiya, Jennifer Welch, Distributed Computing: Fundamentals, Simulations and Advanced Topics, McGraw-Hill.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (VII)	Branch: Computer Science & Engineering
Subject: Cyber Security	Course Code:
Total / Minimum-Pass Marks (End Semester Exam): 100 / 35	L: 1 T: 1 P: 0 Credits: 2
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Course Objectives

1. To Create cyber security awareness and to understand principles of web security
2. To understand key terms and concepts in cyber law, intellectual property and cyber crimes, trademarks and domain theft.
3. To make attentive to students about possible hacking and threats in this communication era.
4. Discuss Issues for creating Security Policy for a Large Organization.

Unit – 1 Cyber Security Fundamentals: Security Concepts: Authentication, Authorization, Non-repudiation, Confidentiality, Integrity, availability. CyberCrimes and Criminals: Definition of cyber-crime, types of cyber-crimes and types of cyber-criminals.

Unit – 2 Cyber attacker Techniques and Motivations: Anti-forensics: Use of proxies, use of tunneling techniques. Fraud techniques: Phishing and malicious mobile code, Rogue antivirus, Click fraud. Threat Infrastructure: Botnets, Fast Flux and advanced fast flux.

Unit – 3 Exploitation: Techniques to gain foothold: Shellcode, Buffer overflows, SQL Injection, Race Conditions, DoS Conditions, Brute force and dictionary attacks. Misdirection, Reconnaissance, and Disruption Methods: Cross-Site Scripting (XSS), Social Engineering, WarXing, DNS Amplification Attacks.

Unit – 4 Information Technology Act 2000: Overview of IT Act 2000, Amendments and Limitations of IT Act, Electronic Governance, Legal Recognition of Electronic Records, Legal Recognition of Digital Signature, Certifying Authorities, Cyber Crime and Offenses, Network Service Providers Liability, Cyber Regulations Appellate Tribunal, Penalties and Adjudication.

Unit – 5 Cyber Law and Related Legislation: Patent Law, Trademark Law, Copyright, Software Copyright or Patented, Domain Names and Copyright disputes, Electronic Data Base and its Protection, IT Act and Civil Procedure Code, IT Act and Criminal Procedural Code, Relevant Sections of Indian Evidence Act, Relevant Sections of Bankers Book Evidence Act, Relevant Sections of Indian Penal Code, Relevant Sections of Reserve Bank of India Act, Law Relating To Employees And Internet, Alternative Dispute Resolution, Online Dispute Resolution (ODR).

Text Books:

1. Cyber Security Essentials, James Graham et al. CRC Press
2. Cyber Laws: Intellectual property & E Commerce Security, Kumar K. Dominant Publisher

Reference Books:

1. Cyber Law Text & Cases, Gerald R. Ferrera, Margo E. K. Reder, CENGAGE LEARNING Publication.
2. Ethics in Information Technology, George W. Reynolds, CENGAGE LEARNING Publication.
3. Cyber Laws & IT Protection, Harish Chander, PHI Publication.
4. Ross J. Anderson. Security Engineering: A Guide to Building Dependable Distributed Systems. John Wiley, New York, NY, 2001.
5. Matt Bishop. Computer Security: Art and Science. Addison Wesley, Boston, MA, 2003.
6. Frank Stajano. Security for Ubiquitous Computing. John Wiley, 2002.

Course Outcomes [After undergoing the course, students will be able to:]

1. Students will be able to acknowledge about the cybercrime, cyber criminal, and intellectual property rights.
2. Encouraging Open Standards.
3. Protection and resilience of Critical Information Infrastructure.
4. To enable effective prevention, investigation and prosecution of cybercrime and enhancement of law enforcement capabilities through appropriate legislative intervention.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (VII)	Branch: Computer Science & Engineering
Subject: Quantum Computing	Course Code: D022733(022)
Total / Minimum-Pass Marks (End Semester Exam): 100 / 35	L: 1 T: 1 P: 0 Credits: 2
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Course Objectives

The objective of this course is to impart necessary knowledge to the learner so that he/she can develop and implement algorithm and write programs using these algorithms.

UNIT-1 Introduction to Quantum Computing: Motivation for studying Quantum Computing, Major players in the industry (IBM, Microsoft, Rigetti, D-Wave etc.), Origin of Quantum Computing. Overview of major concepts in Quantum Computing: Qubits and multi-qubits states, Bra-ket notation, Bloch Sphere representation, Quantum Superposition, Quantum Entanglement

UNIT-2 Math Foundation for Quantum Computing Matrix Algebra: basis vectors and orthogonality, inner product and Hilbert spaces, matrices and tensors, unitary operators and projectors, Dirac notation, Eigen values and Eigen vectors.

UNIT-3 Building Blocks for Quantum Program Architecture of a Quantum Computing platform. Details of q-bit system of information representation: Bloch Sphere, Multi-qubits States, Quantum superposition of qubits (valid and invalid superposition), Quantum Entanglement, Useful states from quantum algorithmic perspective
e.g. Bell State, Operation on qubits: Measuring and transforming using gates, Quantum Logic gates and Circuit: Pauli, Hadamard, phase shift, controlled gates, Ising, Deutsch, swap etc.

UNIT-4 Programming model for a Quantum Computing Program: Steps performed on classical computer, Steps performed on Quantum Computer, Moving data between bits and qubits.

UNIT-5 Quantum Algorithms: Basic techniques exploited by quantum algorithms: Amplitude amplification, Quantum Fourier Transform, Phase Kick-back, Quantum Phase estimation, Quantum Walks, Major Algorithms: Shor's Algorithm, Grover's Algorithm, Deutsch's Algorithm, Deutsch-Jozsa Algorithm. OSS Toolkits for implementing Quantum program: IBM quantum experience Microsoft Q, Rigetti PyQuil (QPU/QVM)

List of Suggested Books:

1. Michael A. Nielsen, "Quantum Computation and Quantum Information", Cambridge University Press.
2. David McMahon, "Quantum Computing Explained", Wiley

Course Outcomes [After undergoing the course, students will be able to:]

1. Explain the working of a Quantum Computing program, its architecture and program model
2. Develop quantum logic gate circuits
3. Develop quantum algorithm
4. Program quantum algorithm on major toolkits

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (VII)	Branch: Computer Science & Engineering
Subject: Distributed Systems	Course Code: D022734(022)
Total / Minimum-Pass Marks (End Semester Exam): 100 / 35	L: 1 T: 1 P: 0 Credits: 2
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Course Objectives

1. Familiarize the students with the basics of distributed computing systems.
2. To introduce the concepts of distributed file systems, shared memory and message passing systems, synchronization and resource management.

Unit I: INTRODUCTION

Definition – Evolution- Goals of distributed systems, system models- Issues in the design of distributed systems- Distributed computing environment.

Unit II: COMMUNICATION

Message Passing – Features and Issues -Synchronization-Buffering – Process Addressing – Failure Handling – Remote procedure call (RPC): Model – Implementation – Stub generation – RPC messages – Marshaling – server Management – Call semantics – communication protocols for RPC-Client server binding – RMI.

Unit III: DISTRIBUTED SHARED MEMORY

Distributed shared memory- Design and implementation issues- Sequential consistency – Release consistency, Process migration Features & Mechanism

Unit IV: SYNCHRONIZATION

Synchronizing physical clocks – Logical clocks – Distributed coordination – Event Ordering – Mutual Exclusion – Deadlock – Election algorithms.

Unit V: DISTRIBUTED FILE SYSTEMS

Introduction – File Models – File accessing, sharing and caching – File Replication – Atomic transactions Case Study HADOOP. : Resource and process management – Task assignment approach – Load balancing approach – Load sharing approach

Text Books:

1. George Colouris, Jean Dollimore and Tim Kindberg, “Distributed Systems – Concepts and Design”, Pearson Education Private Limited, New Delhi, 2001
2. Pradeep K Sinha, “Distributed Operating Systems: Concepts and Design”, Prentice Hall of India, New Delhi, 2003.

Reference Books:

1. Gerard Tel, “Introduction to Distributed algorithms”, Cambridge University Press, USA, 2000.
2. Andrzej Goscinski, “Distributed Operating Systems, the logical Design”, Addison Wesley Publishing Company, USA, 1991.
3. Tanenbaum, “Modern Operating Systems”, Prentice Hall of India, New Delhi, 1999.
4. Patrick Naughton and Herbert Schildt, “Java 2- The Complete Reference”, Tata McGraw Hill, New Delhi, 2007.

Course Outcomes [After undergoing the course, students will be able to:]

1. Verify and analyze the time complexity of the algorithms related to distributed computing.
2. Design and develop various algorithms for problems in distributed computing
3. Compare various resource allocation strategies.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (VII)	Branch: Computer Science & Engineering
Subject: Distributing System	Course Code: D022734(022)
Total / Minimum-Pass Marks (End Semester Exam): 100 / 35	L: 3 T: 1 P: 0 Credits: 4
Class Tests & Assignments to be conducted: 2 each	Duration (End Semester Exam): 03 Hours

Course Objectives

3. Familiarize the students with the basics of distributed computing systems.
4. To introduce the concepts of distributed file systems, shared memory and message passing systems, synchronization and resource management.

Unit I: INTRODUCTION

Definition – Evolution- Goals of distributed systems, system models- Issues in the design of distributed systems- Distributed computing environment.

Unit II: COMMUNICATION

Message Passing – Features and Issues -Synchronization-Buffering – Process Addressing – Failure Handling – Remote procedure call (RPC): Model – Implementation – Stub generation – RPC messages – Marshaling – server Management – Call semantics – communication protocols for RPC-Client server binding – RMI.

Unit III: DISTRIBUTED SHARED MEMORY

Distributed shared memory- Design and implementation issues- Sequential consistency – Release consistency, Process migration Features & Mechanism

Unit IV: SYNCHRONIZATION

Synchronizing physical clocks – Logical clocks – Distributed coordination – Event Ordering – Mutual Exclusion – Deadlock – Election algorithms.

Unit V: DISTRIBUTED FILE SYSTEMS

Introduction – File Models – File accessing, sharing and caching – File Replication – Atomic transactions Case Study HADOOP. : Resource and process management – Task assignment approach – Load balancing approach – Load sharing approach

Text Books:

3. George Colours, Jean Dollimore and Tim Kindberg, “Distributed Systems – Concepts and Design”, Pearson Education Private Limited, New Delhi, 2001
4. Pradeep K Sinha, “Distributed Operating Systems: Concepts and Design”, Prentice Hall of India, New Delhi, 2003.

Reference Books:

5. Gerard Tel, “Introduction to Distributed algorithms”, Cambridge University Press, USA, 2000.
6. Andrzej Goscinski, “Distributed Operating Systems, the logical Design”, Addison Wesley Publishing Company, USA, 1991.
7. Tanenbaum, “Modern Operating Systems”, Prentice Hall of India, New Delhi, 1999.
8. Patrick Naughton and Herbert Schildt, “Java 2- The Complete Reference”, Tata McGraw Hill, New Delhi, 2007.

Course Outcomes [After undergoing the course, students will be able to:]

1. Verify and analyze the time complexity of the algorithms related to distributed computing.
2. Design and develop various algorithms for problems in distributed computing
3. Compare various resource allocation strategies.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (VII)	Branch: Computer Science & Engineering
Subject: Machine Learning Lab	Course Code: D022721(022)
Total / Minimum-Pass Marks (End Semester Exam): 40 / 20	L: 0 T: 0 P: 2 Credits: 1

Course Objectives

1. To be able to use Numpy along with Matplotlib for visual representation of data.
 2. To be able to create a Supervised Learning models in Python.
 3. To be able to create an Un-Supervised Learning models in Python.
 4. To be able to implement Artificial Neural Network in Python.
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1. Write programs to understand the use of Matplotlib for Simple Interactive Chart, Set the Properties of the Plot, matplotlib and NumPy.
 2. Write programs to understand the use of Matplotlib for Working with Multiple Figures and Axes, Adding Text, Adding a Grid and Adding a Legend.
 3. Write programs to understand the use of Matplotlib for Working with Line Chart, Histogram, Bar Chart, Pie Charts.
 4. Write a program in Python to implement Linear Regression for house price prediction. (Data Source: <https://forge.scilab.org/index.php/p/rdataset/source/file/master/csv/MASS/Boston>).
 5. Write a program in Python to implement K Nearest Neighbor classifier for diabetes classification. (Data Source: <https://www.kaggle.com/uciml/pima-indians-diabetes-database/data>).
 6. Build a Naive Bayes model in Python totackle a spam classification problem. (Data Source: (<https://www.kaggle.com/uciml/sms-spam-collectiondataset/downloads/spam.csv/1>)).
 7. Write a Python code to tackle a multi-class classification problem where the challenge is to classify wine into three types using Decision Tree.(DataSource: <https://gist.github.com/tijptjik/9408623/archive/b237fa5848349a14a14e5d4107dc7897c21951f5.zip>).
 8. Write a program in Python to implement Support Vector Machine for diabetes classification. (Data Source: <https://www.kaggle.com/uciml/pima-indians-diabetes-database/data>).
 9. Demonstrate the application of Artificial Neural Network using Python.

Recommended Books:

1. Hands On MachineLearning With Python– John Anderson, AI Sciences LLC.
2. Python for Data Analysis, Wes McKinney, O'Reilly.

Course Outcomes [After undergoing the course, students will be able to:]

1. ApplyNumpy along with Matplotlib for visual analysis of data.
2. Apply Supervised Learning models for problem solving.
3. Apply Un-Supervised Learning models for problem solving.
4. Apply Artificial Neural Network for problem solving.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (VII)	Branch: Computer Science & Engineering
Subject: Internet and Web Technology Lab	Course Code: D022722(022)
Total / Minimum-Pass Marks (End Semester Exam): 40 / 20	L: 0 T: 0 P: 2 Credits: 1

Course Objectives:

- The objective of this lab is to develop an ability to design and implement static and dynamic website
- To learn how to create a simple & advanced web page using HTML along with the usage of style sheets, lists, creation of tables with borders, padding and colors.
- To get acquainted with JavaScript procedures and usage of regular expressions in JavaScript.

Expt-1: Develop and demonstrate a XHTML document that illustrates the use external style sheet, ordered list, table, borders, padding, color, and the tag.

Expt-2: Develop and demonstrate a XHTML file that includes JavaScript for the following problems:

- a) Input: A number *n* obtained using prompt
Output: The first *n* Fibonacci numbers
- b) Input: A number *n* obtained using prompt
Output: A table of numbers from 1 to *n* and their squares using alert

Expt-

3: Develop and demonstrate a XHTML file that includes JavaScript script that uses functions for the following problems:

- a) Parameter: A string
Output: The position in the string of the left-most vowel
- b) Parameter: A number
Output: The number with its digits in the reverse order

Guideline:

1. Declare the script tag as text/javascript in the beginning of the <body> of HTML program
2. Get the number to be reversed from the user using prompt()
3. Validate input number (should be a positive number between 0 to 9) using the regular expression `"/^[0-9]+$/` and alert the user for invalid input using alert()
4. Reverse the number using modulus operation.
5. Use `math.floor(number/10)` to get the floor of number after division (used for reversing)
6. Display the reversed string using alert()

Expt-4(a): Develop and demonstrate, using JavaScript, a XHTML document that collects the USN (the valid format is: A digit from 1 to 4 followed by two upper-case characters followed by two digits followed by two upper-case characters followed by three digits; no embedded spaces allowed) of the user. Event handler must be included for the form element that collects this information to validate the input. Messages in the alert windows must be produced when errors are detected.

(b): Modify the above program to get the current semester also (restricted to be a number from 1 to 8)

Expt-5: Develop and demonstrate, using JavaScript script, a XHTML document that contains three short paragraphs of text, stacked on top of each other, with only enough of each showing so that the mouse cursor can be placed over some part of them. When the cursor is placed over the exposed part of any paragraph, it should rise to the top to become completely visible.

Expt-6: Design an XML document to store information about a student in an engineering college affiliated to CSVTU.

- The information must include Enrolment, Univ_Roll, Name, Name of the College, Branch, Year of Joining, and e-mail id.
- Create sample data for 10 students. Create a CSS stylesheet and use it to display the document.

Expt-7: (a) Write a Perl program to display various Server Information like Server Name, Server Software, Server protocol, CGI Revision etc.

(b) Write a Perl program to accept UNIX command from a HTML form and to display the output of the command executed.

Expt-8: Write a Perl program to accept the User Name and display a greeting message randomly chosen from a list of 4 greeting messages.

Expt-9: Write a Perl program to display a digital clock which displays the current time of the server

Expt-10: Write a Perl program to insert name and age information entered by the user into a table created using MySQL and to display the current contents of this table.

EXPT-11: Write a PHP/.Net program to store current date-time in a COOKIE and display the "Last visited on date-time on the web page upon reopening of the same page.

EXPT-12: Write a PHP/.Net program to store page views count in SESSION, to increment the count on each refresh, and to show the count on web page.

EXPT-13: Create a XHTML form with Name, Address Line 1, Address Line 2, and E-mail text fields. On submitting, store the values in MySQL table. Retrieve and display the data based on Name.

EXPT-14: Using PHP/.Net develop a program to accept book information viz. Accession number, title, authors, edition and publisher from a web page and store the information in a database and to search for a book with the title specified by the user and to display the search results with proper headings.

EXPT-

15: Using PHP/.Net Technology develop an online portal of an online Bookstore. The pages should resemble www.amazon.com the website should consist the following pages.

- Homepage
- Registration and user Login
- User Profile Page
- Books catalog
- Shopping Cart
- Payment By credit card
- Order Confirmation

Validate the Registration, user login, user profile and payment by credit card pages using JavaScript.

Course outcome:

At the end of the course, students should be able to:

- Design and implement dynamic websites with good aesthetic sense of designing and latest technical know-how's.
- Have a good grounding of Web Application Terminologies, Internet Tools, E-Commerce and other web services

Program / Semester: B.Tech (VII)	Branch: Computer Science & Engineering
Subject: Project (Phase I)	Course Code: D022723(022)
Total / Minimum-Pass Marks (End Semester Exam): 60/40	L: 0 T: 0 P: 6 Credits: 3

Guideline for Allocation of project:

1. Information regarding broad area must be made available to the students well in advance (may be during previous semester).
2. Information must cover following parameters.
 - I. Broad area: Subject or expertise/application area.
 - II. Required skills: Knowledge of subject(s), software, tools & other characteristics.
 - III. Type of project: Hardware, software, design, survey, study based etc.
 - IV. Guide available: Name of Guide (S) from Department & Institute.
 - V. Other related information depending upon specific branch & institute.
3. It is also recommended to give proper counseling to pick up suitable project.
4. Students must get chance to select projects as per their choice or decided mutually between students and department faculty (HoD) concern.
5. One project group must contain maximum four students, however students can do project individually but it should be approved by department.
6. Compiled list of projects must be submitted to the University within 25 days of start of semester.
7. Compiled list may contain following parameters.

Monitoring of project:

1. It is recommended to give projects as per the specializations of existing faculty of the department instead of outside person/agency.
2. Project must be allocated, developed and monitored by department / institution itself, but not by outside agencies.
3. Regular review by guide is recommended to ensure development & contribution of students.

Internal Evaluation & Submission of project:

1. Evaluation of project would be as per the examination scheme of the University, which is based on internal as well as external evaluation.
2. Internal assessment requires submission of project report for getting approved by the concern authority.
However printing and binding would be as per the conventional format.
3. Evaluation will be based on live demonstration / presentation and Viva.
4. Final submission of project is expected as,
 - Submission of a copy to the University,
 - One copy to the Institution central library,
 - One copy to the department.

External Evaluation:

External assessment of project would be like conduction of practical exams of University, and must be executed as per the norms of practical exams.

NOTE: Completion of Project outside the department/Institution should not be encouraged.

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (VII)	Branch: Computer Science & Engineering
Subject: Industrial Training	Course Code: D022724(022)
Total Marks (Internal Assessment):20	L: 0 T:0 P: 2 Credit(s): 1
Internal Assessments to be conducted: 02	Duration (End Semester Exam): NA

COURSE OBJECTIVES:

1. To gain experience of working as an engineering professional, including the technical application of engineering knowledge.
2. To experience the discipline of working in a professional organisation and multidisciplinary team.
3. To develop technical, interpersonal and communication skills.

Course Outcomes

On completion of this component of curriculum, the students will be able to

1. Apply engineering knowledge in solving real-life problems.
2. Attain new skills and be aware of the state-of-art in engineering disciplines of their own interest.
3. Get exposure to real-life-working environment & practices, and to attain the professionalisms.
4. Work with multi-tasking professionals and multidisciplinary team.
5. Prepare a technical report, to improve presentation and other soft skills.

Course Content

Exposure to real life problems at various reputed industries engaged in areas of Computer Science and Engineering

Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

Program / Semester: B.Tech (VII)	Branch: Computer Science & Engineering
Subject: Universal Human Values 2	Course Code: D000701(046)
Total Marks (Internal Assessment):10	L: 0 T:0 P: 2 Credit(s): 0
Internal Assessments to be conducted: 02	Duration (End Semester Exam): NA

Course Objectives

1. To help the students appreciate the essential complementarity between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity, which are the core aspirations of all human beings
2. To facilitate the development of a Holistic perspective among students towards life and profession as well as towards happiness and prosperity based on a correct understanding of the Human reality and the rest of Existence. Such a holistic perspective forms the basis of Universal Human Values and movement towards value-based living in a natural way
3. To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct, trustful and mutually fulfilling human behavior and mutually enriching interaction with Nature Thus, this course is intended to provide a much needed orientational input in value education to the young enquiring minds.

UNIT 1: Course Introduction - Need, Basic Guidelines, Content and Process for Value Education- Understanding the need, basic guidelines, content and process for Value Education, Self Exploration-what is it? - its content and process; 'Natural Acceptance' and Experiential Validation- as the mechanism for selfexploration, Continuous Happiness and Prosperity- A look at basic Human Aspirations, Right understanding, Relationship and Physical Facilities- the basic requirements for fulfillment of aspirations of every human being with their correct priority, Understanding Happiness and Prosperity correctly- A critical appraisal of the current scenario 6. Method to fulfill the above human aspirations: understanding and living in harmony at various levels

UNIT 2: Understanding Harmony in the Human Being –

Harmony in Myself!, Understanding human being as a co-existence of the sentient 'I' and the material 'Body', Understanding the needs of Self ('I') and 'Body' - Sukh and Suvidha, Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer), Understanding the characteristics and activities of 'I' and harmony in 'I', Understanding the harmony of I with the Body: Sanyam and Swasthya; correct appraisal of Physical needs, meaning of Prosperity in detail, Programs to ensure Sanyam and Swasthya - Practice Exercises and Case Studies will be taken up in Practice Sessions.

UNIT 3: Understanding Harmony in the Family and Society-

Harmony in Human Human Relationship Understanding Harmony in the family – the basic unit of human interaction, Understanding values in human-human relationship; meaning of Nyaya and program for its fulfillment to ensure Ubhay-tripti; Trust (Vishwas) and Respect (Samman) as the foundational values of relationship, Understanding the meaning of Vishwas; Difference between intention and competence, Recognition of Human-Human Relationship, Recognition of feelings in relationship, Established Values and Expressed Values in Relationship, interrelatedness of feelings and their fulfillment, Expression of feelings, Types of relationship and their purpose, mutual evaluation in relationship, meaning of justice in relationship, Justice leading to culture, civilization and Human Conduct.

UNIT 4: Understanding Harmony in the Nature and Existence –

A comprehensive understanding (knowledge) about the existence, Nature being included; the need and process of inner evolution (through self-exploration, self awareness and self-evaluation), Whole existence as Co-existence, Understanding the harmony in the Nature, Interconnectedness and mutual fulfillment among the four orders of nature recyclability and self-regulation in nature, Understanding Existence as Co-existence (Sah-astitva) of mutually interacting units in all-pervasive space, Holistic perception of harmony at all levels of existence - Practice Exercises and Case Studies will be taken up in Practice Sessions.

UNIT 5: Professional Ethics-

Value based Life and Profession, Professional Ethics and Right Understanding, Competence in Professional Ethics, Issues in Professional Ethics – The Current Scenario, Vision for Holistic Technologies, Production System and Management Models.

Text Books:

1. R R Gaur, R Sangal, G P Bagaria, 2009, A Foundation Course in Human Values and Professional Ethics.
2. A.N Tripathy, New Age International Publishers, 2003..

Reference Books:

1. Ivan Illich, 1974, Energy & Equity, The Trinity Press, Worcester, and Harper Collins, USA
2. E.F. Schumacher, 1973, Small is Beautiful: a study of economics as if people mattered, Blond & Briggs, Britain.
3. Sussan George, 1976, How the Other Half Dies, Penguin Press. Reprinted 1986, 1991
4. Donella H. Meadows, Dennis L. Meadows, Jorgen Randers, William W. Behrens III, 1972, Limits to Growth – Club of Rome's report, Universe Books.
5. Bajpai. B. L , , New Royal Book Co, Lucknow, Reprinted, 2004

Course Outcomes [After undergoing the course, students will be able to:]

1. Understand the significance of value inputs in a classroom, distinguish between values and skills, understand the need, basic guidelines, content and process of value education, explore the meaning of happiness and prosperity and do a correct appraisal of the current scenario in the society
2. Distinguish between the Self and the Body, understand the meaning of Harmony in the Self the Co-existence of Self and Body.
3. Understand the value of harmonious relationship based on trust, respect and other naturally acceptable feelings in human-human relationships and explore their role in ensuring a harmonious society
4. Understand the harmony in nature and existence, and work out their mutually fulfilling participation in the nature.
5. Distinguish between ethical and unethical practices, and start working out the strategy to actualize a harmonious environment wherever they work.

Name of the Program: Bachelor of Technology

Semester: B. Tech – 7th

Subject: Universal Human values 2

Total Marks in End Semester Exam:

Minimum number of Class Tests: 2

Branch: Computer Science Engg.

Course Code: D000701(046)

L: T: P: 2 Credits: 0

Course Objective(s):

- Development of a holistic perspective based on self- exploration about themselves (human being), family, society and nature/existence.
- Understanding (or developing clarity) of the harmony in the human being, family, society and nature/existence
- Strengthening of self-reflection.
- Development of commitment and courage to act.

UNIT-I Introduction- Need, Basic Guidelines, Content and Process for Value Education

- Purpose and motivation for the course, recapitulation from Universal Human Values-I.
- Self-Exploration–what is it? - Its content and process; ‘Natural Acceptance’ and Experiential Validation- as the process for self-exploration.
- Continuous Happiness and Prosperity- A look at basic Human Aspirations
- Right understanding, Relationship and Physical Facility- the basic requirements for fulfillment of aspirations of every human being with their correct priority.
- Understanding Happiness and Prosperity correctly- A critical appraisal of the current scenario
- Method to fulfill the above human aspirations: understanding and living in harmony at various levels.
- Include practice sessions to discuss natural acceptance in human being as the innate acceptance for living with responsibility (living in relationship, harmony and co-existence) rather than as arbitrariness in choice based on liking-disliking.

UNIT-II Understanding Harmony in the Human Being - Harmony in Myself!

- Understanding the needs of Self (‘I’) and ‘Body’ - happiness and physical facility.
- Understanding the Body as an instrument of ‘I’ (I being the doer, seer and enjoyer).
- Understanding the characteristics and activities of ‘I’ and harmony in ‘I’.
- Understanding the harmony of I with the Body: Sanyam and Health; correct appraisal of Physical needs, meaning of Prosperity in detail.
- Programs to ensure Sanyam and Health.
- Include practice sessions to discuss the role others have played in making material goods available to me. Identifying from one’s own life.
- Differentiate between prosperity and accumulation. Discuss program for ensuring health vs dealing with disease

UNIT-III Understanding Harmony in the Family and Society- Harmony in Human-Human Relationship

- Understanding values in human-human relationship; meaning of Justice (nine universal values in relationships) and program for its fulfilment to ensure mutual happiness; Trust and Respect as the foundational values of relationship
- Understanding the meaning of Trust; Difference between intention and competence
- Understanding the meaning of Respect, Difference between respect and differentiation; the other salient values in relationship
- Understanding the harmony in the society (society being an extension of family): Resolution, Prosperity, fearlessness (trust) and co-existence as comprehensive Human Goals
- Visualizing a universal harmonious order in society- Undivided Society, Universal Order- from family to world family.

- Include practice sessions to reflect on relationships in family, hostel and institute as extended family, real life examples, teacher-student relationship, goal of education etc. Gratitude as a universal value in relationships. Discuss with scenarios. Elicit examples from students' lives.

UNIT-IV Understanding Harmony in the Nature and Existence - Whole existence as Coexistence

- Understanding the harmony in the Nature
- Interconnectedness and mutual fulfilment among the four orders of nature- recyclability and self-regulation in nature.
- Understanding Existence as Co-existence of mutually interacting units in all-pervasive space.
- Holistic perception of harmony at all levels of existence.
- Include practice sessions to discuss human being as cause of imbalance in nature (film "Home" can be used), pollution, depletion of resources and role of technology etc.

UNIT-V Implications of the above Holistic Understanding of Harmony on Professional Ethics

- Natural acceptance of human values
- Definitiveness of Ethical Human Conduct
- Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order
- Competence in professional ethics: a. Ability to utilize the professional competence for augmenting universal human order b. Ability to identify the scope and characteristics of people friendly and eco-friendly production systems, c. Ability to identify and develop appropriate technologies and management patterns for above production systems.
- Case studies of typical holistic technologies, management models and production systems
- Strategy for transition from the present state to Universal Human Order:
 - At the level of individual: as socially and ecologically responsible engineers, technologists and managers
 - At the level of society: as mutually enriching institutions and organizations
- Include practice Exercises and Case Studies will be taken up in Practice (tutorial) Sessions e.g. to discuss the conduct as an engineer or scientist etc.

Text Books:

1. Human Values and Professional Ethics by R R Gaur, R Sangal, G P Bagaria, Excel Books, New Delhi, 2010
2. Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3. Jeevan Vidya: EkParichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.

Reference Books:

1. The Story of Stuff (Book).
2. The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi.
3. Small is Beautiful - E. F Schumacher.

Course Outcome:

After completion of course, student should be able to

- To become more aware of themselves, and their surroundings (family, society, nature); they would become more responsible in life, and in handling problems with sustainable solutions, while keeping human relationships and human nature in mind.
- They would have better critical ability. They would also become sensitive to their commitment towards what they have understood (human values, human relationship and human society). It is hoped that they would be able to apply what they have learnt to their own self in different day-to- day settings in real life, at least a beginning would be made in this direction.